

D-2.3 Petri Dish

D-2.4 Burette — 50-ml capacity.

D-2.5 Pipette — 10-ml capacity.

D-2.6 Exposure Cell — A specially designed stainless steel cell comprising of two cylindrical chambers having cross section area 100 cm² and 10 cm height which may be coupled without leakage with the help of a silicone rubber seal (see Fig. 1). The test fabric is sandwiched between these two chambers and tightened with nuts and bolts. Both the cells shall have in and out vents controlled through precision needle valves for introduction and withdrawal of air for analysis.

D-3. REAGENTS

D-3.1 Aluminium Phosphide Tablet — Each tablet weighing approximately 3 g and containing minimum 56 percent (*m/m*) active ingredients.

D-3.2 Mercuric Chloride Solution — 10 percent (*m/v*) solution of mercuric chloride in water.

D-3.3 Standard Iodine Solution — 0.1 N iodine solution. Dissolve 12.7 g of iodine in water containing 30 g of potassium iodide and make up the volume to 1 litre in volumetric flask.

D-3.4 Standard Sodium Thiosulphate Solution — 0.1 N sodium thiosulphate solution. Dissolve 24.8 g of sodium thiosulphate crystals (Na₂S₂O₃·5H₂O) in water and make up the volume to 1 litre in volumetric flask.

D-3.5 Starch Indicator Solution — Make a paste of 1 g of soluble starch with a little water. The paste is then added to 100 ml of boiling water with constant stirring. The resulting solution is boiled for another minute and then cooled.

D-4. STANDARD ATMOSPHERIC CONDITIONS FOR TESTING

D-4.1 Test shall be carried out in an atmosphere of 65 ± 5 percent relative humidity and at a temperature of 27 ± 2°C.

D-5. PROCEDURE

D-5.1 Estimation of Phosphine Concentration in Aspirator Bottle — Two to three aluminium phosphide tablets are kept in aspirator bottle. The bottle is made air-tight and filled with air having relative humidity more than 60 percent. After the equilibration period of 24 h, 50 ml phosphine laden air is drawn from this bottle through a three-way stopcock into a gas-tight syringe and then slowly bubbled into 20 ml of 10 percent mercuric chloride solution. The precipitate formed is filtered, washed and then dissolved in 10 ml of 0.1 N iodine solution. Titrate this iodine solution

against standard sodium thiosulphate solution using starch solution as indicator.

D-5.1.1 Carry out a blank by using all reagents in the same quantities except phosphine gas.

D-5.1.2 Calculation — Calculate the concentration of phosphine in enclosed space of aspirator bottle as follows:

$$\text{Phosphine, mg/l (c)} = \frac{0.425 \times V \times N \times 1\,000}{0.1 \times 50}$$

where

V = final volume of standard sodium thiosulphate solution in ml, used for titration of iodine solution, after correction for blank; and

N = normality of standard sodium thiosulphate solution.

D-5.1.2.1 Estimate the volume of phosphine required to produce a concentration of 3 mg/l in the upper chamber of apparatus as follows:

$$\text{Volume of phosphine required (ml)} = \frac{1\,000 \times 3}{c}$$

where

c = concentration of phosphine in mg/l (see 5.1.2).

D-5.2 Estimation of Escaped Phosphine — 15 ml of 10 percent mercuric chloride solution is kept in a wide petri dish in the lower chamber. The test fabric is fitted in between the two chambers which are clamped tightly. The fabric shall be properly and uniformly stretched between the two chambers. The upper chamber is partially evacuated by the gas-tight syringe and required volume of air containing phosphine to produce 3 mg/l concentration, as calculated in D-5.1.2.1, is introduced into it. After 24 h of exposure the test chambers are opened and the amount of phosphine permeated in the lower chamber (which forms precipitate with mercuric chloride solution kept in petri dish) is estimated following the procedure of dissolving precipitate in iodine solution and titrating it against standard sodium thiosulphate solution as given in D-5.1.

D-5.2.1 Carry out a blank by using all reagents in the same quantities except phosphine gas.

D-5.2.2 Calculation — Calculate the concentration of phosphine (mg) permeated in the lower chamber in 24 h as follows:

$$\text{Concentration of phosphine, mg (Z)} = \frac{0.425 \times V \times N}{0.1}$$

where

V = final volume of standard sodium thiosulphate solution in ml, used for titration of iodine solution after correction for blank; and

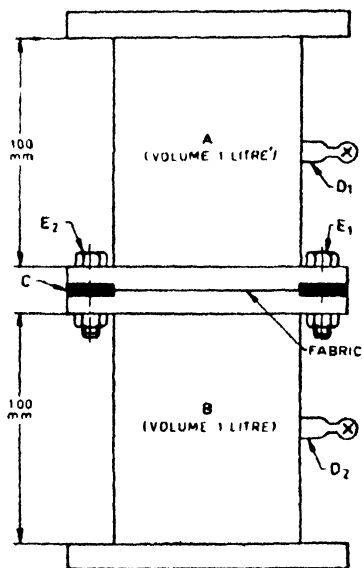
N = normality of standard sodium thiosulphate solution.

D-5.2.2.1 Calculate the percentage escaped phosphine in 24 h as follows:

$$\text{Escaped phosphine, percent} = \frac{Z \times 100}{3}$$

where

Z = concentration of phosphine, in mg, permeated in lower chamber (see 5.2.2).



A and B = Stainless steel chambers

C = Silicone rubber seal

D_1, D_2 = Inlets for fumigants

E_1 and E_2 = Clamping nuts and bolts

FIG. 1 APPARATUS FOR CHECKING RETENTION OF FUMIGANTS

(PCDC 16)